

Critical Analysis of the Ever-Changing Estimates of the Hubble Constant

Increasing variability in estimates of the Hubble Constant (H_0) has exacerbated the Hubble tension. This paper investigates the reasons behind such discrepancies and presents a comparative study of the methods of H_0 determination. Several experiments were performed using the Cosmic Distance Ladder (CDL) to explore the cumulative effect of systematic errors on the final H_0 value. A figure of $73 \pm 8 \text{ km s}^{-1} \text{ Mpc}^{-1}$ was estimated at distances of up to 25 Mpc, which agrees with the SH_0ES and Hubble Space Telescope (HST) measurements. Further up the ladder, the systematic errors were found to become statistically insignificant in comparison to the distance, but a drastic decrease in the value of H_0 was found. The best estimate of $52 \pm 3 \text{ km s}^{-1} \text{ Mpc}^{-1}$ was obtained at 250 Mpc, hinting at the anisotropic behaviour of H_0 and the possibility of non-uniform cosmic expansion.

1. Introduction

Edwin Hubble's revolutionary discovery of cosmic expansion in 1929 [1] gave impetus to interesting new research in the field of cosmology. The Hubble Constant (H_0) quantifies the universal expansion rate and is fundamental in understanding the structure and characteristics of the current cosmological model [2]. Knowledge of the composition of the universe depends on a well-grounded value of H_0 which can help theorize the fate of the cosmos and give insight into the early universe. It is the key to estimating the cosmic age [2] and calculating the baryonic density that resulted from primordial nucleosynthesis during the early cosmological era [2-5]. A deep understanding of the energy and matter density of the universe assists in interpreting its geometry, the behaviour of its expansion and the elements responsible for it [2,6,7]. H_0 can be calculated using the recessional velocities and

distances of galaxies sufficiently far such that their peculiar velocities become insignificant compared to the Hubble Flow velocity [8,9]. Analysis of stellar spectra provides an efficient route to determining the redshifts, and hence the recessional velocities [2], while the determination of distance relies on the inverse relationship between light intensity and the square of distance [2]. The absolute and apparent magnitudes of astronomical objects can be related by the distance-modulus equation [10]:

$$m - M = 5 \log\left(\frac{d}{10}\right) \quad (1)$$

where m = apparent brightness, M = absolute brightness and d = distance. Hence, if an object has a known absolute magnitude which can be compared to its apparent brightness, its distance can be estimated [2]. Such objects form the cosmic distance ladder (CDL), a series of methods assisting in the determination of

galactic distances (Figure 1). Owing to their unique attributes, the Cepheids, Type Ia Supernovae (SNe Ia) and Red Giant Branch (RGB) Stars are typically used for this task. The luminosities of the Cepheids exhibit oscillatory behaviour first discovered by Henrietta Swan Leavitt in 1908 [2,11]. She defined the Leavitt Law that linearises the period-luminosity relationship (PLR) of the Cepheids as per the equation [12]:

$$M = a(\log P - 1) + b \quad (2)$$

Where M = absolute magnitude and P = Period of the Cepheids. The Cepheids are also useful

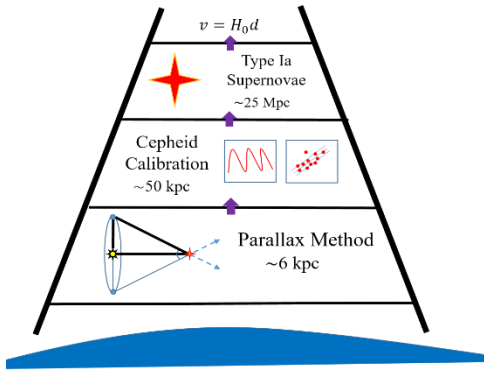


Figure 1. The Cosmic Distance Ladder is a series of methods that assists in determination of galactic distances. Each method is effective to a certain distance.

as secondary distance indicators, allowing calibration of other astronomical objects for measurements at greater distances [13]. The luminosity of an SNe Ia is known to peak at an absolute brightness of -19.3 ± 0.3 [14]. These stars can be used with Cepheids, RGB stars, and the baryon acoustic oscillation (BAO) method to determine H_0 [7,15]. The red giants experience a dramatic decrease in absolute brightness once a sufficient core temperature triggers helium burning. This decrease appears

as a discontinuity in the colour-magnitude diagram at predictable values that can be calibrated [16,17]. The experiment in this paper focused on employing multiple methods of distance determination of target galaxies to compare and analyse the varying estimates of H_0 . Once these distances were obtained, the gradient of the linear correlation between them and the recessional velocities was evaluated, giving the value of H_0 . The experiment is divided into “Setups” to show the effect of adopting a new procedure. The obtained results are compared to the findings of the Hubble Space Telescope (HST) Key Project [13], the primary source of data for this experiment, and the Supernova H_0 for the Equation of State (SH_0ES) Project [18] that had a similar methodology as explained in Section 2. Common sources of systematic errors are examined in Section 4, followed by a case-by-case evaluation of each setup. Improvements and mitigation strategies follow in Section 5.

2. Methodology

In Setup 1, the estimate of H_0 was made using photometric data of the Cepheids obtained by the HST. A sinusoidal approximation to the data allowed for determining the period of their luminosities. The Leavitt Law (Eq. 2) was then used to calculate the absolute magnitude, and hence the distances to the host galaxies (Eq. 1). Setup 2 involved distance measurements by taking the inverse of trigonometric parallaxes recorded by the European Space Agency’s Gaia

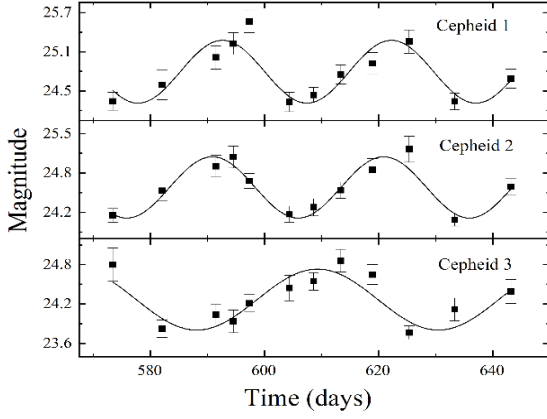


Figure 2. A sinusoidal approximation of the form $y = y_0 + A \sin \left[\frac{\pi(x-x_c)}{w} \right]$ is made to the period-luminosity relationship of the Cepheids; y_0 gives the average apparent brightness and $2w$ gives the period of the oscillation.

Mission, GDR3 [19] (Figure 3). This method significantly reduced the systematic errors introduced by sinusoidal approximations, reddening effects and zero-point calibrations [15]. This is further discussed in Section 4. The experiment was extended to reproduce the Leavitt Law using absolute magnitudes evaluated by geometrical measurements of distances. The “new” Leavitt Law could then be used to calibrate galaxies at greater distances. Reliance on the Cepheid Variables restricted the scope of observation to up to 25 Mpc. Use of brighter standard candles, such as SNe Ia, allowed to expand the scope to at least 2.5% of the observable universe. In Setup 3, using the photometric data of SNe Ia retrieved from the Asiago Supernova Catalogue (Version 2008) [20], galactic distances of up to 250 Mpc could be estimated. Finally, the Cepheid and SNe Ia data were aggregated to work out the best estimate of H_0 . In all Hubble plots, an additional

point of (0, 0) served as the point of observation stationary in its rest frame and around which the universe was assumed to expand uniformly.

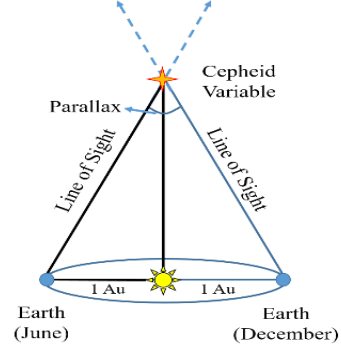


Figure 3. The apparent displacement of stars when viewed from different perspectives allows the application of trigonometry to estimate the distances.

3. Results

The values of H_0 were found to be different using each approach. In Setup 1, a value of $61 \pm 5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ was estimated with an uncertainty of 8%. In Setup 2, the Leavitt Law was graphically reproduced using parallax measurements of the Cepheids, giving the

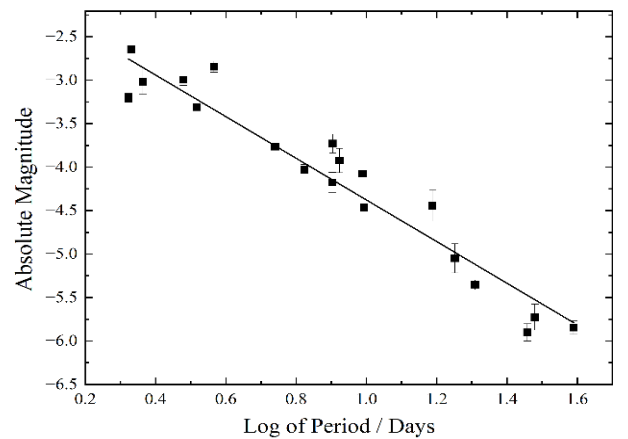


Figure 4. The Leavitt Law linearises the period-luminosity relationship of the Cepheids. The equation of the form $M = A \log P + C$ is used. The gradient, $A = -2.39 \pm 0.16$ and the intercept, $C = -1.98 \pm 0.13$

gradient of the plot, A , as -2.39 ± 0.16 and the intercept C as -1.98 ± 0.13 (Figure 4). With these values, H_0 was evaluated as $73 \pm 8 \text{ km s}^{-1} \text{ Mpc}^{-1}$ at up to 25 Mpc from the point of observation, showing an uncertainty of 11% and hence an increase in errors with increasing distances. Extending observations to 250 Mpc from the point of observation by involving photometric measurements of SNe Ia, the value significantly decreased to $49 \pm 4 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Piecing together all the available data gave the best estimate of $52 \pm 3 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Figure 5), which was used to calculate the age of the Universe as $19 \pm 1 \text{ Gyr}$.

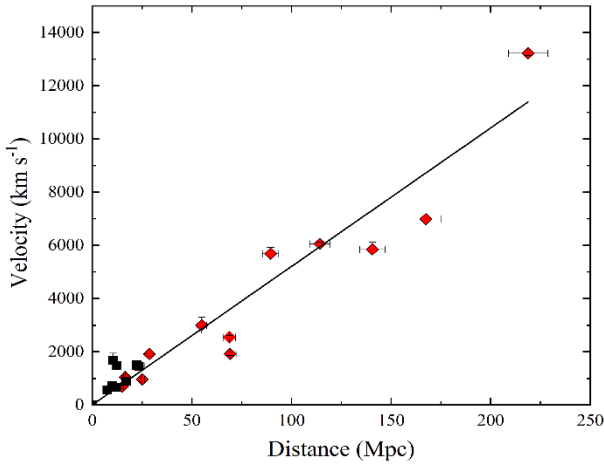


Figure 5. Hubble Plot of the aggregate data of Cepheids (■) and SNe Ia (♦). The Cepheid samples are localised in a small area whereas the SNe Ia samples are more spread-out. The error bars in the Cepheid measurements become less significant at large distances.

4. Discussion

The severity of the Hubble tension can be noticed from the extent of variability in the results. Extragalactic distances are often estimated by using the CDL as illustrated in

Figure 1. The systematic errors introduced in each rung of the ladder propagate and cumulate through to the end [21]. The galactic redshifts were determined by aligning only the Ca H and Ca K absorption lines, contrary to using a complete comparison spectrum. The uncertainties were noted by visually observing the width of the troughs in the spectrum, giving inaccurate values. Research shows that the HST estimated H_0 displays an anisotropic behaviour [22]. It is therefore safe to assume that a cosmological bias may be introduced by sampling galaxies in a specified direction and disregarding the rest of the neighbourhood. This should not matter assuming the universal expansion rate is uniform. However, as shown later, this may not be the case. Finally, a limited number of galaxies were used in the evaluation from those catalogued by the HST. The experiment discussed measures distances of up to 2.5% of the observable universe, making the sample non-representative. The initial estimate of H_0 is found to be discrepant by 5.3σ from the SH_0ES value of $73.52 \pm 1.62 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [3], and by 4.2σ from the HST value of $72 \pm 8 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [13]. As can be identified from Figure 2, the sinusoidal approximation fits well with only one of the three light curves (i.e. Cepheid 2), showing that it was unreasonable to assume all Cepheid light curves to have a sinusoidal shape. Most of the Cepheids have asymmetric light curves [23] and rely upon techniques such as Fourier Decomposition for the determination of their

periods [24]. For this setup, the effects of reddening were also not corrected. The value of H_0 obtained from Setup 2 is in strong agreement with the SH_0ES and HST values. Substituting periodic approximations and multiple calculations with direct geometrical measurement mitigated many of the systematic errors. Using accurate periods and

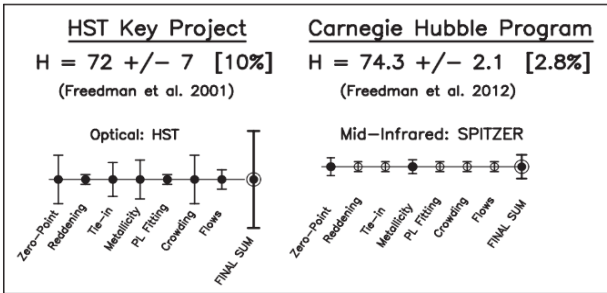


Figure 6. The systematic error budget of the HST Key Project compared to the Carnegie Hubble Program that relies on the TRGB method for H_0 determination (reproduced from [15]).

reddening-corrected data from the HST, the required constants for the Leavitt Law were evaluated and accurate measurements for extragalactic Cepheids were obtained. Although an overall improvement from Setup 1, the strong agreement with the HST values can be deceptive. A recent analysis of the HST findings reveals large zero-point calibration errors in Cepheid photometry [15]. Figure 6 shows the systematic error budget of the HST Key Project. By using SNe Ia as standard candles, these issues could be evaded. A significant decrease in the values of H_0 is observed as target distances increase. The best estimate of H_0 found using aggregate data varies by about 15σ from the SH_0ES and HST measurements. The

uncertainty drops to 6% as reliance on the Cepheids is minimized. The drop in value further demonstrates the anisotropy of H_0 . It also appears that the expansion rate of the universe is non-uniform. The introduction of the Λ CDM cosmological model and the role of dark matter and dark energy in the expansion of the universe entertains the possibility of an accelerating or decelerating universe [3,25,26], corroborating these conclusions. Expectedly, a decreased expansion rate sets a higher age for the universe than 13 Gyr as currently accepted [2].

5. Further Work

The tip of the red-giant branch (TRGB) method provides a viable substitute for the Cepheids as it can be applied to all galaxies [17]. TRGBs do not require reddening correction as colour selection at the time of sampling is sufficient to obtain unbiased data [10]. Being relatively isolated, TRGBs do not suffer from cosmological biases introduced by crowding or blending effects [10]. A novel method for H_0 determination relies on mapping the variation in remnant radiation from the Big Bang [7]. Independent of the CDL, it gives a much lower estimate of $67.8 \pm 0.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$, varying at 4.4σ from the HST and SH_0ES measurements [7]. The results are found to agree with those obtained from the BAO and SNe Ia methods [7]. The discrepancies may result from reliance on the physics of the early universe which still requires further investigation [6,7,27].

6. Conclusion

In this paper, the H_0 estimates were found to vary upon changing methods of measurement. At increasing distances, the H_0 value was found to decrease, with the best estimate of $52 \pm 3 \text{ km s}^{-1} \text{ Mpc}^{-1}$ differing at 15σ from HST and SH_0ES measurements. This value sets an older cosmic age than conventionally estimated and defines a non-uniform expansion rate for the universe in agreement with newer cosmological models such as Λ CDM. A comparative study of different methods followed, showing that minimal reliance on Cepheids and the CDL resulted in smaller systematic errors. Further work still needs to be done before the enigma surrounding the characteristics of the universe can be solved.

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